FINITE GENERATION OF THE UNITS MODULO CONSTANTS UNDER ANY ONE OF THREE ADDITIONAL HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Finite generation of units for the algebraically closed case and integral closure of the ground field. Stacks Project, Varieties, Lemma lemma-units-general-algebraically-closed (tag 09NM) and Lemma lemma-integral-closure-ground-field (tag 09NN). The first is applied to the geometric component and the second identifies the integral closure with a field.
- Irreducible components of a base change along a separably closed field. Stacks Project, Varieties, Lemma lemma-separably-closed-field-irreducible-components (tag 09NV). Ensures that the geometric component has the same irreducible components, so the algebraically closed case applies.

Source reference. Original reference 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01WM: sources/stacks-chow/tags.txt, source lines 2487–2487. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04KZ: sources/stacks-chow/tags.txt, source lines 5990–5990. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 038H: sources/stacks-chow/tags.txt, source lines 4236–4236. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04L6: sources/stacks-chow/tags.txt, source lines 5997–5997. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04MI: sources/stacks-chow/tags.txt, source lines 6044–6044. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let k be a field. Let X be a scheme over k. Assume that X is locally of finite type over k, connected, reduced, and has finitely many irreducible*

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

components. Then $\mathcal{O}(X)^*/k^*$ is a finitely generated abelian group if in addition to the conditions above at least one of the following conditions is satisfied:

(1) *the integral closure of k in $\Gamma(X, \mathcal{O}_X)$ is k,*
(2) *X has a k-rational point, or*
(3) *X is geometrically integral.*

Proof. Let $\overline{k}$ be an algebraic closure of k. Let Y be a connected component of $(X_{\overline{k}})_{red}$. Note that the canonical morphism $p : Y \to X$ is open (by Morphisms, Lemma 0383) and closed (by Morphisms, Lemma 01WM). Hence $p(Y) = X$ as X was assumed connected. In particular, as X is reduced this implies $\mathcal{O}(X) \subset \mathcal{O}(Y)$. By Lemma 04KZ we see that Y has finitely many irreducible components (the field $\overline{k}$ in the lemma is the separable algebraic closure, but that this doesn't matter as the two base changes $X_{\overline{k}}$ have the same irreducible components by Lemma 038H). Thus Lemma 04L6 applies to Y. This implies that if $\mathcal{O}(X)^*/k^*$ is not a finitely generated abelian group, then there exist elements $f \in \mathcal{O}(X)$, $f \notin k$ which map to an element of $\overline{k}$ via the map $\mathcal{O}(X) \to \mathcal{O}(Y)$. In this case f is algebraic over k, hence integral over k. Thus, if condition (1) holds, then this cannot happen. To finish the proof we show that conditions (2) and (3) imply (1).

Let $k \subset k' \subset \Gamma(X, \mathcal{O}_X)$ be the integral closure of k in $\Gamma(X, \mathcal{O}_X)$. By Lemma 04MI we see that k' is a field. If $e : \mathrm{Spec}(k) \to X$ is a k-rational point, then $e^\sharp : \Gamma(X, \mathcal{O}_X) \to k$ is a section to the inclusion map $k \to \Gamma(X, \mathcal{O}_X)$. In particular the restriction of $e^\sharp$ to k' is a field map $k' \to k$ over k, which clearly shows that (2) implies (1).

If the integral closure k' of k in $\Gamma(X, \mathcal{O}_X)$ is not trivial, then we see that X is either not geometrically connected (if k'/k is not purely inseparable) or that X is not geometrically reduced (if k'/k is nontrivial purely inseparable). Details omitted. Hence (3) implies (1). □